

1. What does the ACID acronym stand for in database transaction management?
 - (A) Atomicity, Consistency, Isolation, Durability
 - (B) Accuracy, Coherence, Integrity, Distribution
 - (C) Availability, Concurrency, Indexing, Durability
 - (D) Asynchronous, Centralized, Immutable, Deterministic

2. Which consensus algorithm is designed to be easily understood and equivalent to Paxos in fault tolerance?
 - (A) Byzantine Agreement
 - (B) Raft Consensus
 - (C) Gossip Protocol
 - (D) Proof of Work

3. What is the fundamental mechanism behind Redis in-memory pub/sub message brokering?
 - (A) Persistent WAL log replication
 - (B) Ephemeral memory-buffered channel broadcasting
 - (C) Distributed LSM disk tree
 - (D) Two-phase commit protocol

4. Examine the following complex network partitioning scenario:
In a 5-node distributed cluster experiencing a network split between nodes {A, B} and {C, D, E}, determine which partition continues accepting writes under majority quorum rules:
 - (A) Partition {A, B} because node A holds the initial leader lease
 - (B) Both partitions continue writes independently and merge via CRDTs

- (C) Partition {C, D, E} because it maintains a strict quorum majority (>50%)
- (D) Neither partition can accept writes until network partition heals

5. State True or False:

Under CAP theorem, a distributed database can achieve both strong consistency and 100% availability during a network partition.

- (A) True
- (B) False

ANSWER KEY:

- 1. A
- 2. B
- 3. B
- 4. C
- 5. B